

Daily Sequences Drill #4 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: *Recurrence Relations · The Characteristic Equation · Closed-Form Solutions.*

Section A Rapid Recognition

1. A recurrence has $a_1 = 2$ and $a_{n+1} = 3a_n - 1$. Find a_2 and a_3 .

Answer: $a_2 = 5, a_3 = 14$

Method: $a_2 = 3(2) - 1 = 5$. $a_3 = 3(5) - 1 = 14$.

Investigate further: This is a first-order recurrence with a constant additive term — exactly the shape C1/C2 solve properly in closed form later in this sheet.

2. A recurrence has $a_1 = 1$ and $a_{n+1} = 2a_n + 1$. Find a_4 .

Answer: 15

Method: $a_2 = 2(1) + 1 = 3$. $a_3 = 2(3) + 1 = 7$. $a_4 = 2(7) + 1 = 15$.

Investigate further: Notice the pattern $1, 3, 7, 15, \dots$: each is one less than a power of 2 ($2, 4, 8, 16$) — worth guessing the closed form $a_n = 2^n - 1$ and checking it.

3. Write down the characteristic equation of $a_n = 5a_{n-1} - 6a_{n-2}$.

Answer: $x^2 = 5x - 6$ (i.e. $x^2 - 5x + 6 = 0$)

Method: Replace a_n, a_{n-1}, a_{n-2} with x^2, x^1, x^0 respectively in $a_n = 5a_{n-1} - 6a_{n-2}$.

Investigate further: This substitution is purely formal — it works because the recurrence is linear with constant coefficients, so a solution of the form $a_n = x^n$ must satisfy this polynomial exactly.

4. Solve $x^2 - 5x + 6 = 0$ for its roots.

Answer: $x = 2, 3$

Method: $x^2 - 5x + 6 = (x - 2)(x - 3) = 0$.

Investigate further: These are the same two numbers as the recurrence's coefficients 5 (their sum) and 6 (their product) — always true by Vieta's formulas on the characteristic equation.

5. A recurrence has characteristic roots 2 and 3. Write down the general form of its closed-form solution.

Answer: $a_n = A \cdot 2^n + B \cdot 3^n$

Method: Two distinct real roots r_1, r_2 give a general solution $a_n = Ar_1^n + Br_2^n$, with A, B determined

by initial conditions.

Investigate further: This is the recurrence-relation analogue of a second-order linear ODE's general solution — two independent “basis” sequences, combined linearly.

6. The recurrence $a_{n+1} = \frac{1}{2}a_n$, $a_1 = 8$ is secretly which type of sequence from an earlier day? Write its closed form.

Answer: A GP; $a_n = 8\left(\frac{1}{2}\right)^{n-1}$

Method: $a_{n+1} = \frac{1}{2}a_n$ is a constant-ratio recurrence, i.e. exactly a geometric sequence with $a = 8$, $r = \frac{1}{2}$ (Day 3's toolkit).

Investigate further: Recognising this instantly, without reaching for a characteristic equation, is the point of B9/C6 later in this sheet.

7. Find the fixed point L of the recurrence $a_{n+1} = 3a_n + 4$ (i.e. solve $L = 3L + 4$).

Answer: $L = -2$

Method: $L = 3L + 4 \Rightarrow -2L = 4 \Rightarrow L = -2$.

Investigate further: L is the value at which the recurrence would stay put forever if it ever landed exactly on it — the fixed point is the anchor the whole shift-to-GP trick (B2, B3, C1, C2) is built around.

8. True or false: every recurrence $a_n = pa_{n-1} + qa_{n-2}$ with real, distinct characteristic roots has a closed form that's a combination of two exponential terms.

Answer: True

Method: This is exactly the general solution form derived in A5: two distinct real roots give a linear combination of two exponential sequences.

Investigate further: This stops being true the moment the roots repeat or go complex — A10 and C4 cover those two other cases.

9. The sequence $a_n = 2^n + 3^n$ satisfies a recurrence $a_n = pa_{n-1} + qa_{n-2}$. Find p and q .

Answer: $p = 5, q = -6$

Method: Since $a_n = 2^n + 3^n$ is built from roots 2 and 3, its characteristic equation is $(x-2)(x-3) = x^2 - 5x + 6 = 0$, i.e. $x^2 = 5x - 6$, giving $a_n = 5a_{n-1} - 6a_{n-2}$.

Investigate further: This is A3/A4 run in reverse — given the roots (or a sequence built from them), reconstruct the recurrence rather than solve one.

10. A recurrence has a repeated characteristic root $r = 2$. Write down the general form of its closed-form solution.

Answer: $a_n = (A + Bn)2^n$

Method: A repeated root r needs a second, genuinely independent solution alongside r^n — multiplying by n (giving nr^n) supplies it, so the general solution is $a_n = (A + Bn)r^n$.

Investigate further: This mirrors what happens with a repeated root of a polynomial more generally — an extra factor of n (or x) is needed to keep the two basis solutions linearly independent.

Section B Manipulation Drills

1. Solve the recurrence $a_n = 5a_{n-1} - 6a_{n-2}$ with $a_1 = 1, a_2 = 4$, for the constants A, B in $a_n = A \cdot 2^n + B \cdot 3^n$.

Answer: $A = -\frac{1}{2}, B = \frac{2}{3}$

Method: $n = 1$: $2A + 3B = 1$. $n = 2$: $4A + 9B = 4$. Doubling the first: $4A + 6B = 2$; subtracting from the second: $3B = 2 \Rightarrow B = \frac{2}{3}$. Then $2A + 2 = 1 \Rightarrow A = -\frac{1}{2}$.

Investigate further: Always check both original conditions after solving: $a_1 = -\frac{1}{2}(2) + \frac{2}{3}(3) = -1 + 2 = 1 \checkmark$, $a_2 = -\frac{1}{2}(4) + \frac{2}{3}(9) = -2 + 6 = 4 \checkmark$.

2. For $a_{n+1} = 3a_n + 4$, the fixed point is $L = -2$. Letting $b_n = a_n - L = a_n + 2$, what type of sequence is (b_n) ?

- A) A GP with ratio 3.
 B) An AP with common difference 3.
 C) A GP with ratio -2 .
 D) Neither.

Answer: A) A GP with ratio 3.

Method: $b_{n+1} = a_{n+1} + 2 = (3a_n + 4) + 2 = 3a_n + 6 = 3(a_n + 2) = 3b_n$ — so (b_n) satisfies a pure geometric recurrence.

Investigate further: This is exactly why the shift $b_n = a_n - L$ works in general: subtracting the fixed point cancels the additive constant, leaving pure multiplication behind.

3. Continuing from B2: given $a_1 = 1$ (so $b_1 = 3$), find a_5 .

- A) 241
 B) 243
 C) 239
 D) 245

Answer: A) 241

Method: $b_1 = a_1 + 2 = 3$, and (b_n) is a GP with ratio 3: $b_n = 3 \cdot 3^{n-1} = 3^n$. So $b_5 = 3^5 = 243$, and $a_5 = b_5 - 2 = 241$.

Investigate further: Always convert back from b_n to a_n at the very end — forgetting the -2 shift-back is the most common slip in this technique.

4. Find the characteristic equation and its roots for $a_n = a_{n-1} + 2a_{n-2}$.

Answer: $x^2 - x - 2 = 0$; roots $x = 2, -1$

Method: $x^2 = x + 2 \Rightarrow x^2 - x - 2 = 0 = (x - 2)(x + 1) \Rightarrow x = 2, -1$.

Investigate further: A negative characteristic root is completely normal — it produces a sign-alternating term $B(-1)^n$ in the closed form, as B5 uses directly.

5. A recurrence has characteristic roots 2 and -1 , with $a_1 = 1, a_2 = 5$. Which closed form fits?

- A) $a_n = 2^n + (-1)^n$
 B) $a_n = 2^n - (-1)^n$
 C) $a_n = 2 \cdot 2^n - (-1)^n$
 D) $a_n = 2^{n-1} + (-1)^{n-1}$

Answer: A) $a_n = 2^n + (-1)^n$

Method: Check $n = 1$: $2 + (-1) = 1 \checkmark$. Check $n = 2$: $4 + 1 = 5 \checkmark$. The other options fail at $n = 1$: B) gives $2 - (-1) = 3$, C) gives $4 + 1 = 5$, D) gives $1 + 1 = 2$ — none match $a_1 = 1$.

Investigate further: With two initial conditions and two unknowns (A, B in $A \cdot 2^n + B \cdot (-1)^n$), the closed form is uniquely determined — always verify a proposed closed form against *both* given terms, not just one.

6. Which of the following recurrences does NOT have a characteristic equation with two distinct real roots?

- A) $a_n = 5a_{n-1} - 6a_{n-2}$
 B) $a_n = 4a_{n-1} - 4a_{n-2}$
 C) $a_n = a_{n-1} + 2a_{n-2}$
 D) $a_n = 3a_{n-1} - 2a_{n-2}$

Answer: B) $a_n = 4a_{n-1} - 4a_{n-2}$

Method: Its characteristic equation is $x^2 - 4x + 4 = (x - 2)^2 = 0$ — a repeated root, discriminant $16 - 16 = 0$. The other three all have positive discriminant (A: $25 - 24 = 1$; C: $1 + 8 = 9$; D: $9 - 8 = 1$), giving distinct real roots.

Investigate further: Computing the discriminant of the characteristic equation directly ($p^2 + 4q$ for $a_n = pa_{n-1} + qa_{n-2}$) is faster than factorising every option by hand.

7. A recurrence has repeated characteristic root $r = 3$, with $a_1 = 6, a_2 = 27$. Using $a_n = (A + Bn)3^n$, find A and B .

- A) $A = 1, B = 1$
 B) $A = 2, B = 1$
 C) $A = 1, B = 2$
 D) $A = 0, B = 2$

Answer: A) $A = 1, B = 1$

Method: $(A + B \cdot 1) \cdot 3^1 = 6 \Rightarrow A + B = 2$. $(A + B \cdot 2) \cdot 3^2 = 27 \Rightarrow A + 2B = 3$. Subtracting: $B = 1$, then $A = 1$.

Investigate further: Double-check: $a_1 = (1 + 1) \cdot 3 = 6 \checkmark$, $a_2 = (1 + 2) \cdot 9 = 27 \checkmark$.

8. Using the closed form from B7, find a_4 .

Answer: 405

Method: $a_n = (1+n)3^n$. $a_4 = (1+4) \cdot 3^4 = 5 \times 81 = 405$.

Investigate further: The repeated-root closed form grows faster than a plain exponential r^n because of the extra factor of n — compare $a_4 = 405$ here to what a plain $c \cdot 3^4 = 81c$ would need ($c = 5$) to match, which is a coincidence of this specific example, not a general shortcut.

9. A recurrence is defined by $a_{n+1} = a_n + 5$, $a_1 = 2$. Is this recurrence's closed form better found using the AP toolkit (Day 1) or the characteristic-equation toolkit (Day 4)?

- A) The AP toolkit — it's simply $a_n = 2 + 5(n-1)$.
- B) The characteristic-equation toolkit — solve $x = x + 5$.
- C) Both give the same closed form, but the characteristic equation is faster.
- D) Neither toolkit applies.

Answer: A) The AP toolkit — it's simply $a_n = 2 + 5(n-1)$.

Method: A constant additive step (+5 every time, no multiplication) is precisely Day 1's arithmetic sequence definition — the characteristic-equation machinery is unnecessary (and its "fixed point" equation $x = x + 5$ has no solution, signalling this recurrence sits outside that method's normal use case).

Investigate further: Recognising when a lighter toolkit already solves the problem is as important as knowing the heavier one — see Top 5 Pattern 4.

10. A recurrence has characteristic equation $x^2 - 6x + 9 = 0$. What is special about its roots?

- A) Two distinct real roots.
- B) A repeated real root.
- C) Two complex roots.
- D) No real roots.

Answer: B) A repeated real root.

Method: $x^2 - 6x + 9 = (x-3)^2 = 0$ — discriminant $36 - 36 = 0$.

Investigate further: This is the same equation structure as B7's recurrence, $a_n = 6a_{n-1} - 9a_{n-2}$ — worth noticing the connection.

Section C Substitution & Structure

1. Which substitution transforms $a_{n+1} = 4a_n - 3$ into a pure geometric recurrence $b_{n+1} = 4b_n$?

- A) $b_n = a_n - 1$
- B) $b_n = a_n + 1$
- C) $b_n = a_n - 3$
- D) $b_n = a_n/4$

Answer: A) $b_n = a_n - 1$

Method: Fixed point: $L = 4L - 3 \Rightarrow -3L = -3 \Rightarrow L = 1$. So $b_n = a_n - 1$, giving $b_{n+1} = a_{n+1} - 1 = (4a_n - 3) - 1 = 4a_n - 4 = 4(a_n - 1) = 4b_n$.

Investigate further: The fixed point of $a_{n+1} = pa_n + c$ is always $L = \frac{c}{1-p}$ (for $p \neq 1$) — worth memorising as a one-line formula rather than resolving $L = pL + c$ from scratch every time.

2. The recurrence $a_{n+1} = 4a_n - 3$ with $a_1 = 2$ has closed form $a_n = 1 + c \cdot 4^{n-1}$. Find c .

- A) 1
B) 2
C) 3
D) 4

Answer: A) 1

Method: At $n = 1$: $a_1 = 1 + c \cdot 4^0 = 1 + c = 2 \Rightarrow c = 1$.

Investigate further: c here is exactly b_1 from the shift-to-GP method (C1) — the two approaches (“closed form with unknown constant” vs. “shift to a GP”) are the same technique in different notation.

3. A recurrence has characteristic roots $r_1 = 5$ and $r_2 = -2$. Using $p = r_1 + r_2$ and $q = -r_1r_2$, write down the recurrence $a_n = pa_{n-1} + qa_{n-2}$.

- A) $a_n = 3a_{n-1} + 10a_{n-2}$
B) $a_n = 3a_{n-1} - 10a_{n-2}$
C) $a_n = -3a_{n-1} + 10a_{n-2}$
D) $a_n = 7a_{n-1} - 10a_{n-2}$

Answer: A) $a_n = 3a_{n-1} + 10a_{n-2}$

Method: $p = 5 + (-2) = 3$. $q = -(5)(-2) = 10$. Check via the characteristic equation directly: $(x - 5)(x + 2) = x^2 - 3x - 10 = 0 \Rightarrow x^2 = 3x + 10 \Rightarrow a_n = 3a_{n-1} + 10a_{n-2}$.

Investigate further: This is A9’s technique in general form — always rebuild the characteristic equation from the roots directly as a check, rather than trusting the $p = r_1 + r_2$, $q = -r_1r_2$ formulas from memory alone.

4. A recurrence has characteristic equation $x^2 - 2x + 5 = 0$. What can be said about its roots?

- A) Two distinct real roots.
B) A repeated real root.
C) Two complex conjugate roots, $1 \pm 2i$.
D) No roots exist.

Answer: C) Two complex conjugate roots, $1 \pm 2i$.

Method: Discriminant $= (-2)^2 - 4(1)(5) = 4 - 20 = -16 < 0$, so the roots are complex: $x = \frac{2 \pm \sqrt{-16}}{2} = \frac{2 \pm 4i}{2} = 1 \pm 2i$.

Investigate further: A negative discriminant doesn't mean "no solution" for a recurrence the way it might for a real-world root-finding problem — the recurrence still has a well-defined (complex-valued, or real-oscillating) closed form, explored further in D3.

5. Which technique is most appropriate for solving $a_{n+1} = 2a_n + 3^n$ (note the non-constant additive term)?

- A) The constant-shift trick ($b_n = a_n - L$ for a fixed L) — works directly.
- B) The homogeneous characteristic-equation method, plus a particular solution of the form $C \cdot 3^n$ added to the general solution.
- C) Treating it as a plain AP.
- D) It cannot be solved by any method covered so far.

Answer: B) The homogeneous method plus a particular solution $C \cdot 3^n$.

Method: The constant-shift trick only cancels a *constant* forcing term — here the forcing term 3^n grows with n , so no fixed constant L can absorb it. The standard fix is to add a particular solution of the same form as the forcing term ($C \cdot 3^n$) to the homogeneous general solution.

Investigate further: This is the recurrence-relation analogue of "guess a particular solution matching the forcing function's form" from differential equations — the same idea, one level more discrete.

6. Which of the following recurrences is secretly equivalent to an arithmetic sequence — Day 1's toolkit is sufficient, no characteristic equation needed?

- A) $a_{n+1} = a_n + 7$
- B) $a_{n+1} = 2a_n$
- C) $a_{n+1} = 2a_n + 7$
- D) $a_n = 5a_{n-1} - 6a_{n-2}$

Answer: A) $a_{n+1} = a_n + 7$

Method: A) is a constant additive step — an AP, Day 1's toolkit exactly. B) is a constant-ratio step — a GP (Day 3). C) needs the shift-to-GP trick (still not a plain AP). D) genuinely needs the full characteristic equation.

Investigate further: This question is the whole sheet's "triage" skill in miniature: always ask whether Day 1 or Day 3's lighter toolkit already applies before reaching for Day 4's heavier machinery.

7. A recurrence $a_n = pa_{n-1} + qa_{n-2}$ has BOTH characteristic roots equal to 1. What does this imply about a_n as $n \rightarrow \infty$ (assuming (a_n) is not identically zero)?

- A) Exponential growth.
- B) Linear growth in n (or constant) — effectively an AP.
- C) The sequence must be periodic.
- D) The sequence must converge to a finite limit.

Answer: B) Linear growth in n (or constant) — effectively an AP.

Method: A repeated root $r = 1$ gives closed form $a_n = (A + Bn)(1)^n = A + Bn$ — exactly linear in n (an AP) if $B \neq 0$, or constant if $B = 0$.

Investigate further: This is a genuinely elegant full-circle result: the “two independent exponential terms” structure of a general second-order recurrence collapses back down to Day 1’s arithmetic sequence exactly when both roots equal 1 — see D4 for the reverse direction (which (p, q) forces this).

8. A recurrence has characteristic roots r and $\frac{1}{r}$ (reciprocal roots, $r \neq \pm 1$). What can be said about p and q in $a_n = pa_{n-1} + qa_{n-2}$?

- A) $q = -1$ always, while $p = r + \frac{1}{r}$ can be any value with $|p| \geq 2$.
 B) $p = -1$ always.
 C) p and q must both equal 1.
 D) No general statement can be made.

Answer: A) $q = -1$ always, $p = r + \frac{1}{r}$ can be any value with $|p| \geq 2$.

Method: $q = -r_1r_2 = -r \cdot \frac{1}{r} = -1$, independent of r . And $p = r + \frac{1}{r}$, which by AM–GM-style reasoning satisfies $|r + \frac{1}{r}| \geq 2$ for any nonzero real r (with equality only at $r = \pm 1$).

Investigate further: The bound $|r + \frac{1}{r}| \geq 2$ is the same inequality that appears throughout the Algebra pillar’s toolkit — a nice example of one pillar’s tool resurfacing to describe a genuinely different structure (reciprocal characteristic roots) here.

Section D Challenge

1. A sequence is defined by $a_1 = 1, a_2 = 2$, and for $n \geq 3$: $a_n = \frac{a_{n-1} + a_{n-2}}{2}$. Find $\lim_{n \rightarrow \infty} a_n$.

Answer: $\frac{5}{3}$

Method: Characteristic equation: $2x^2 = x + 1 \Rightarrow 2x^2 - x - 1 = 0 = (2x + 1)(x - 1) \Rightarrow x = 1, -\frac{1}{2}$. General solution: $a_n = A + B(-\frac{1}{2})^n$. Using $a_1 = 1$: $A - \frac{B}{2} = 1$. Using $a_2 = 2$: $A + \frac{B}{4} = 2$. Subtracting: $\frac{3B}{4} = 1 \Rightarrow B = \frac{4}{3}$, then $A = 1 + \frac{2}{3} = \frac{5}{3}$. As $n \rightarrow \infty$, $(-\frac{1}{2})^n \rightarrow 0$, so $a_n \rightarrow A = \frac{5}{3}$.

Investigate further: Numerically: 1, 2, 1.5, 1.75, 1.625, 1.6875, ... — oscillating and visibly closing in on $1.6 = \frac{5}{3}$. A root of exactly 1 in an otherwise-decaying recurrence always signals convergence to a finite, computable limit, never to 0.

2. A recurrence satisfies $a_{n+1} = 6a_n - 9a_{n-1}$ (for $n \geq 2$), with $a_1 = 1, a_2 = 6$. Using the repeated-root closed form $a_n = (A + Bn)3^n$, find a_5 .

- A) 405
 B) 243
 C) 324
 D) 486

Answer: A) 405

Method: Characteristic equation $x^2 - 6x + 9 = (x - 3)^2 = 0$ — repeated root 3. Solving $a_n = (A + Bn)3^n$:

$n = 1$: $(A + B)(3) = 1 \Rightarrow A + B = \frac{1}{3}$. $n = 2$: $(A + 2B)(9) = 6 \Rightarrow A + 2B = \frac{2}{3}$. Subtracting: $B = \frac{1}{3}$, so $A = 0$. Thus $a_n = \frac{n}{3} \cdot 3^n = n \cdot 3^{n-1}$. $a_5 = 5 \times 3^4 = 5 \times 81 = 405$.

Investigate further: Check the pattern directly: $a_n = n \cdot 3^{n-1}$ gives 1, 6, 27, 108, 405, ... — verify $a_3 = 6a_2 - 9a_1 = 6(6) - 9(1) = 27$ ✓ matches the recurrence.

3. A recurrence has characteristic equation $x^2 + 4 = 0$ (i.e. $a_n = -4a_{n-2}$). What is true about its behaviour?

- A) The sequence oscillates in sign while its magnitude grows like 2^n — consistent with complex roots of magnitude 2.
- B) The sequence converges to 0.
- C) The sequence is eventually constant.
- D) The sequence is purely real-valued growth with no oscillation.

Answer: A) The sequence oscillates in sign while its magnitude grows like 2^n .

Method: $x^2 = -4 \Rightarrow x = \pm 2i$, purely imaginary roots of magnitude 2. Testing directly with $a_1 = 1, a_2 = 0$: $a_3 = -4a_1 = -4$, $a_4 = -4a_2 = 0$, $a_5 = -4a_3 = 16$, $a_6 = -4a_4 = 0$, $a_7 = -4a_5 = -64$ — the nonzero terms alternate sign and grow by a factor of 4 every two steps (i.e. magnitude 2 per step), exactly matching complex roots of modulus 2.

Investigate further: This is the general pattern for complex characteristic roots $re^{i\theta}$: the real sequence's magnitude grows like r^n , while it oscillates with a “period” related to θ — purely imaginary roots ($\theta = 90$) give the fastest possible sign-oscillation, a full cycle every 4 steps.

4. For which of the following (p, q) pairs does $a_n = pa_{n-1} + qa_{n-2}$ produce a closed form that reduces exactly to Day 1's arithmetic-sequence form (i.e. a_n linear in n), for suitable initial conditions?

- A) $(p, q) = (2, -1)$
- B) $(p, q) = (1, 1)$
- C) $(p, q) = (-2, 1)$
- D) $(p, q) = (3, -2)$

Answer: A) $(p, q) = (2, -1)$

Method: Linear-in- n behaviour needs a repeated root of exactly 1: $(x - 1)^2 = x^2 - 2x + 1 = 0$, giving $p = 2, q = -1$ (matching $x^2 - px - q = 0$). Checking D: $(p, q) = (3, -2)$ gives $x^2 - 3x + 2 = (x - 1)(x - 2) = 0$ — roots 1, 2, distinct, so the general solution $A + B \cdot 2^n$ is linear in n only in the degenerate special case $B = 0$, not for suitable general initial conditions.

Investigate further: This is C7's result read backwards — given the desired long-term behaviour (linear growth), work out which characteristic equation, and hence which (p, q) , produces it.

5. A recurrence has characteristic roots r_1, r_2 with $|r_1| < 1$ and $|r_2| < 1$. What happens to a_n as $n \rightarrow \infty$, regardless of initial conditions?

- A) $a_n \rightarrow 0$ always.
- B) a_n diverges to infinity.

- C) a_n oscillates forever without settling.
 D) It depends on the initial conditions.

Answer: A) $a_n \rightarrow 0$ always.

Method: $a_n = Ar_1^n + Br_2^n$ (or the repeated-root analogue). Since $|r_1| < 1$ and $|r_2| < 1$, both $r_1^n \rightarrow 0$ and $r_2^n \rightarrow 0$ as $n \rightarrow \infty$, regardless of the constants A, B (which are fixed by the initial conditions but don't affect whether the limit is 0).

Investigate further: This is the recurrence-relation version of Day 3's convergence condition $|r| < 1$ for a GP — both roots need to be inside the unit circle in magnitude, not just one. Tomorrow's sheet studies exactly this kind of boundedness and convergence behaviour in general.

Top 5 Patterns — Worksheet #4

- Characteristic equation: turn $a_n = pa_{n-1} + qa_{n-2}$ into $x^2 - px - q = 0$.** The entire toolkit starts here. (see A3, A4, A9, B4, C3)
- Distinct real roots $\rightarrow a_n = Ar_1^n + Br_2^n$; repeated root $\rightarrow a_n = (A + Bn)r^n$.** Always check the discriminant first to know which shape applies. (see A5, A10, B5, B7, C7)
- First-order recurrences $a_{n+1} = pa_n + c$ shift to a pure GP via the fixed point $L = \frac{c}{1-p}$.** Let $b_n = a_n - L$, solve the GP, shift back. (see A7, B2, B3, C1, C2)
- Recognise the degenerate cases first.** Constant-difference recurrences are Day 1's AP; constant-ratio recurrences are Day 3's GP — don't reach for the characteristic equation when a lighter toolkit already solves it. (see A6, B9, C6)
- Complex characteristic roots signal oscillation with exponentially changing magnitude,** not simple convergence or divergence. (see C4, D3)

Common Traps — Worksheet #4

- Forgetting to check for a repeated root before using the two-distinct-roots closed form.** A zero discriminant needs $(A + Bn)r^n$ instead. (see B6, B10)
- Mixing up the sign convention when reading p, q off from given roots.** Remember $x^2 - px - q = 0$, not $x^2 + px + q = 0$. (see C3, D4)
- Applying the constant-shift trick to a recurrence with a non-constant forcing term.** $b_n = a_n - L$ only cancels a fixed constant — it does nothing against a term like 3^n . (see C5)
- Assuming every recurrence converges to a finite limit.** Convergence needs *every* characteristic root to have magnitude strictly less than 1. (see D5)
- Solving for A, B using only one initial condition.** A second-order recurrence's closed form has two free constants and needs two equations to pin them both down. (see B1, B5, B7)

"Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised."

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths